

Surprisal as the Drive of Subjective Time: A Gauge-Constrained Taxonomy and Its Decisive Test

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Abstract

A companion formal note shows that any reparameterization-invariant readout of the proper-time trajectory alone collapses to an endpoint functional, so interior sensitivity to subjective time requires a path-distinguishing, τ -intrinsic *drive*; that result is silent on *which* drive. We propose a principled candidate—the *surprisal* of experience, $s = -\log p$, so that felt duration is accumulated information, $\Delta P = k \int s d\tau$ —gauge-invariant by construction. The paper is built around the single measurement that decides it—a behavioural contrast that confirms surprisal, selects a rival, or falsifies surprisal outright. From this single currency we derive a battery of falsifiable predictions: a saturating, finite-ceiling age curve (with a late-life slowdown variant), an order effect in which clustered events read longest, a clean separation from contextual-change accounts, and Weber–Fechner intensity scaling. The evidence that supports it: the age prediction holds against digitized lifespan data (a plateau to age 94, ruling out proportional and unbounded laws); in the open Blursday database ($N = 2,376$) greater *subjective* confinement predicts a markedly slower momentary passage of time ($p \approx 4 \times 10^{-12}$); and a back-heavy convergence asymmetry emerges in embodied agents that never read the drive. Where the data already strain it: the *objective*-mobility version of the locality test is null, and the retrospective arm is wrong-signed—greater confinement *expands* rather than compresses remembered duration—so what survives is the narrower claim that felt isolation, not spatial extent, slows momentary time. Because these consistency checks do not adjudicate surprisal against its nearest rival, we close with the decisive contrast that does: a single behavioural test on which surprisal predicts a clustered event-burst reads *longest* while a contextual-change account predicts *shortest*. A single currency ties subjective time to its drivers, and we are explicit throughout about where the data strain it.

1 Introduction

The felt speed of time is not fixed: it stretches under novelty and compresses under routine, and it changes systematically across the lifespan. A companion formal note establishes a structural constraint on *any* model of this phenomenon: if the readout is invariant under reparameterization of physical time and depends on the proper-time trajectory alone, it is determined by the endpoints of that trajectory (with or without internal memory). Interior sensitivity—the dependence of felt duration on *how* an interval was filled—therefore demands a path-distinguishing, τ -intrinsic input [1].

That theorem is a gate, not a mechanism: it says what a drive must look like, not what it is. Existing accounts supply drives operationally (a pacemaker rate, an attentional gate, a count of contextual changes [2, 3, 4]) but leave open *why* those particular quantities. We propose that the drive is the *surprisal* of experience, $s = -\log p(\text{event})$ under the observer’s predictive model—the information each moment carries. Surprisal is a single, principled currency; this paper asks what it

predicts and tests one prediction directly. More than a model, it is a model paired with the single measurement that decides it: §6 sets out a behavioural contrast whose outcome confirms surprisal, selects a named rival, or falsifies surprisal outright—the experiment the paper is built to provoke.

2 An information-theoretic principle for subjective time

We state the framework as a principle and develop its consequences; the psychology of later sections is its instantiation, not its foundation.

2.1 The principle

Let a stimulus stream $x(\tau)$ be defined on proper time τ and let the observer hold a predictive model $\mathcal{M}_\tau$ whose precision sharpens with experience. With the surprisal (self-information) $s(\tau) = -\log p(x(\tau) \mid \mathcal{M}_\tau)$, we posit that the subjective duration of an interval equals the information registered over it:

$$\Delta P = k \int_{\tau_0}^{\tau_T} s(\tau) d\tau = k \mathbb{I}_{[\tau_0, \tau_T]}, \quad s(\tau) = -\log p(x(\tau) \mid \mathcal{M}_\tau). \quad (1)$$

Perceived duration is accumulated information; the felt *rate* of time is an information density. The scale k fixes units (bits to felt time) and is a single global constant, not a per-prediction knob: every falsifiable consequence below is a *ratio* (the finite age ceiling, §2.3), an *ordering* (the reorder effect, §3), or a *log-slope* (Weber–Fechner, §3), each invariant under $k \rightarrow ck$, so k cannot be tuned to rescue any one of them. Two facets follow, with opposite sign, and they are the two empirically established modes of time experience. A *remembered* interval that registered little information reads as short — it “went fast” (retrospective speed $\propto 1/\bar{s}$); a *passing* interval is felt through the rate of information currently being realized (momentary speed $\propto s$). This is the classical retrospective/momentary (“holiday paradox”) split, here a single principle viewed two ways.

The principle is not free-standing: surprisal $-\log p$ is the quantity self-organizing agents are argued to minimize under the free-energy principle [5], and duration models in which perceived time accumulates salient perceptual change or surprise-gated episodic memories are already established [6, 7]—indeed the holiday-paradox reading above is theirs. We do not claim the accumulation idea as novel. Our contribution is to (i) place it under the reparameterization-invariance constraint of the companion note, which selects surprisal (a τ -intrinsic functional) over coordinate-rate drives; (ii) show that a *single* information currency then also yields the age ceiling (§2.3), Weber–Fechner scaling (§3) and a spatial-locality *prediction* (§2.4)—whose status the data qualify (§5); and (iii) put that locality prediction to a direct test. That “information” here is a physical, not metaphorical, quantity is underwritten by the information–thermodynamics correspondence [8, 9].

2.2 Gauge invariance (corollary)

Because $\mathbb{I}$ in (1) is a functional of the proper-time data $(\tau, x(\tau))$ alone, it is invariant under any reparameterization $t \rightarrow t'$ of physical time — it never reads a coordinate-time quantity. This is exactly the constraint the companion note proves necessary, and it separates the surprisal drive from coordinate-surprise foils (which read $d\tau/dt$). Numerically (Fig. 1) the drive’s response to a battery of time-warps converges to zero under refinement while a coordinate-dependent observer converges to a nonzero floor.

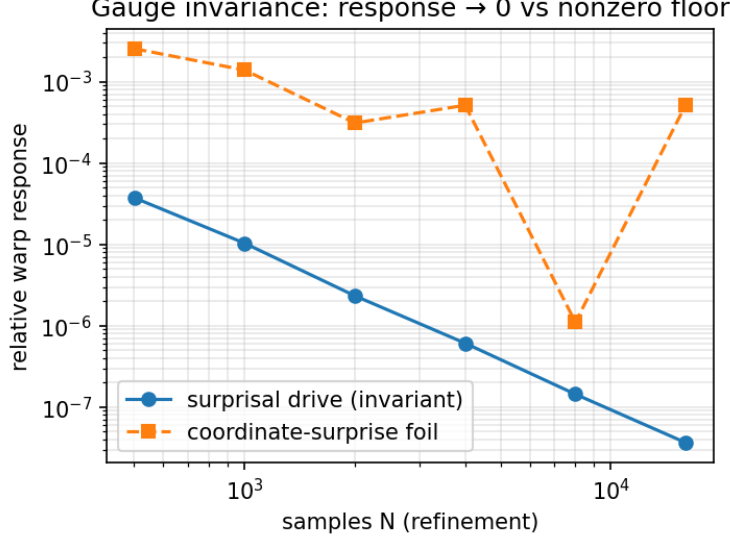


Figure 1: Gauge invariance: the surprisal drive’s warp response vanishes under refinement (pure discretization); a coordinate-surprise foil plateaus at a nonzero floor.

Which reparameterization (and which it is not). The invariance at issue—and the only one the companion note requires—is under reparameterization of *physical time*, $t \rightarrow t'$. It holds for (1) whatever the form of p , because ΔP integrates the scalar density $s(\tau)$ against the invariant measure $d\tau$ along the fixed proper-time trajectory, so a smooth relabeling of the time coordinate leaves it unchanged; the numerical warp test (Fig. 1) certifies exactly this. A *distinct* symmetry—reparameterization of the observable itself, $x \mapsto \varphi(x)$ —is neither claimed nor needed, and for a continuous density it does not hold: $-\log p(x)$ then acquires a Jacobian term $-\log |\varphi'|$. Surprisal is therefore defined relative to a fixed stimulus encoding (the observable’s natural measure). The discrete-token instantiation of §4 is automatically invariant on the observable axis (a relabeling only permutes a probability mass function); in the continuous case, if observable coordinate-freeness is also wanted, the drive generalizes to a *relative* surprisal—a Kullback–Leibler divergence against a fixed reference measure—which is coordinate-free by construction. This is a generalization, not a correction: none of the predictions below depends on it.

2.3 Missing information, Fisher information, and a finite age ceiling

Let the environment emit $x \sim \mathcal{N}(\mu, \sigma^2)$ with μ a learnable parameter and σ^2 irreducible noise. After experience a (observation count) the observer’s posterior over μ has variance $v(a) = (\lambda_0 + a \mathcal{I}_F)^{-1}$, where $\mathcal{I}_F = \sigma^{-2}$ is the per-observation Fisher information: *Fisher information accumulates linearly with experience, and missing information about the learnable structure falls as $\frac{1}{2} \log v(a) \sim -\frac{1}{2} \log a$* . The one-step predictive variance is $\sigma^2 + v(a)$, so the expected surprisal rate is the predictive entropy

$$r(a) = \frac{1}{2} \log(2\pi e [\sigma^2 + v(a)]) \xrightarrow{a \rightarrow \infty} \frac{1}{2} \log(2\pi e \sigma^2) \equiv r_\infty. \quad (2)$$

The rate does not vanish: the learnable missing information disappears ($v \rightarrow 0$) but the *irreducible* missing information set by σ^2 remains. Hence the retrospective felt speed $s(a) \propto r(\text{ref})/r(a)$ rises and **saturates at a finite ceiling** $r(\text{ref})/r_\infty$. The saturating age curve is therefore not assumed: it is the signature of Fisher-information accumulation against a noise floor. Strict proportionality has no floor, predicts unbounded speed-up, and is the law the lifespan data reject (§3).

2.4 Entropy capacity and locality

A locality affording $W(\ell)$ distinguishable states bounds the entropy of the observation distribution by $\log W(\ell)$ (maximum entropy at fixed support [8]), and hence bounds the achievable information rate, $r \leq \log W(\ell)$ per unit time. The realized momentary rate is $\rho \log W(\ell)$ with an engagement fraction $\rho \in [0, 1]$: *locality sets the ceiling on information rate; attention/boredom sets the fraction realized*. Shrinking the locality (confinement, $W \downarrow$) lowers the ceiling, and monotony lowers ρ ; both reduce accrued information and—by (1)—slow the felt passage of time. This is a state-counting (Boltzmann) bound: “the size of a region bounds the information experienced in it” already at the level of accessible microstates, with no appeal to anything exotic. We flag in advance that the test in §5 supports the *momentary slowing* but *not* the capacity (W) reading: an objective-mobility proxy for W is null, implicating the engagement term ρ rather than spatial capacity.

2.5 Outlook: information and area (conjecture, not load-bearing)

The strongest “information $\leq$ area” statement in physics is the Bekenstein/holographic bound, $S \leq A/4\ell_P^2$ [10]. It is tempting to read perceived time — accumulated information along a worldline — as ultimately area-bounded. We record this only as a conjecture: the bound is not operative here (information registered by an observer sits dozens of orders of magnitude below it), so it constrains nothing measurable. It would graduate from analogy to physics only given a *derived* relation between worldline-accumulated surprisal and a geometric/area quantity of the relevant region, which we do not have. Nothing in §§2.3–2.4 or below depends on it.

3 Predictions

The surprisal drive is not free of structure; it makes specific, falsifiable predictions, each implemented as an additive simulation.

Age: a finite ceiling, and a late-life slowdown. As the predictive model absorbs the learnable structure of the world, surprisal per unit time declines toward an irreducible floor (the world is never fully predictable), so a fixed interval carries less surprisal with age and feels faster—and the speed-up *saturates* at a finite ceiling rather than growing without bound (Fig. 2b). Digitized lifespan data [11] show exactly this: the felt speed of the “last ten years” rises steeply to age ~ 50 and then plateaus to age 94. Fit against the candidate family, the saturating law wins decisively while strict proportionality ($s \propto \text{age}$) over-predicts threefold and is rejected; a logarithmic law is admissible in range but, lacking a ceiling, diverges from the saturating law only in the oldest tail (Fig. 2a). A variant in which surprisal *rises again* in very old age (disrupted routine, lost mastery) yields a peak-then-slowdown curve, reproducing reports that momentary time slows after ~ 75 [12].

Temporal reorder: clustered events read longest. For a fixed set of events, only their arrangement is varied. The surprisal drive predicts that a dense mid-interval burst—maximally surprising against a settled baseline—reads as the longest, robustly across encoding (Fig. 3). The front-versus-back contrast is encoding-dependent and does not cleanly adjudicate the model; clustered-versus-dispersed is its signature.

Separation from contextual-change. A dense burst is maximally *surprising* (surprisal $\rightarrow$ long) yet occupies *few distinct contexts* (contextual-change $\rightarrow$ short): the two accounts are anti-correlated

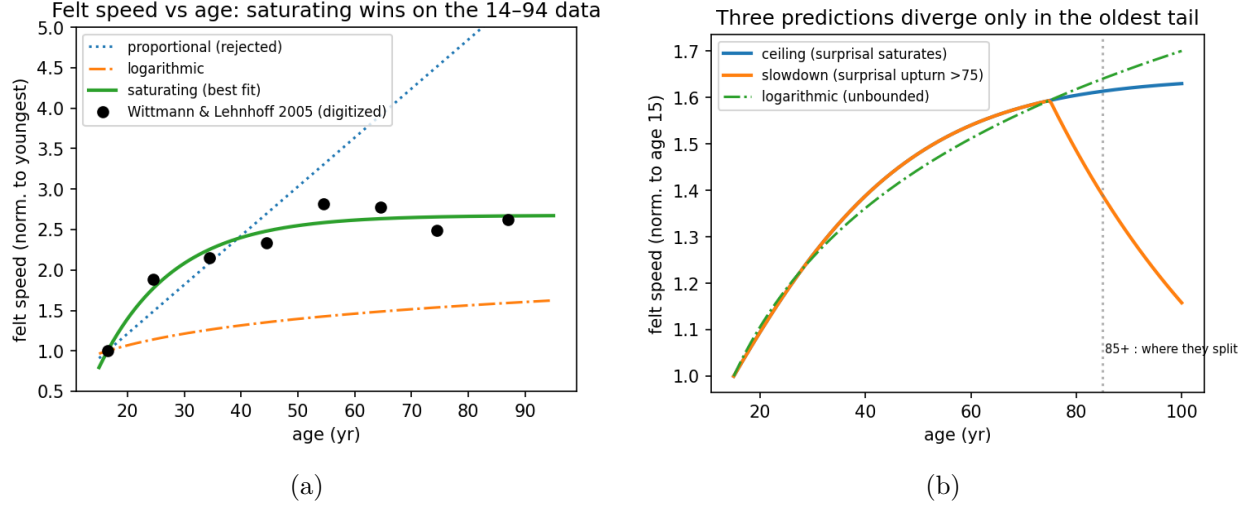


Figure 2: (a) Candidate felt-speed laws against digitized lifespan data [11]: saturating wins; proportional and logarithmic miss. (b) Ceiling, slowdown, and logarithmic laws diverge only in the oldest tail.

on clustering and thus cleanly distinguishable by a single question —does a dense burst after calm feel long or short? (Fig. 3).

Weber–Fechner from the same currency. If stimulus intensity carries a scale-free prior $p(I) \propto 1/I$ (the maximum-entropy prior for a positive scale variable), then surprisal $-\log p(I) = \log I + \text{const}$ is exactly Fechner’s logarithmic magnitude, and its differential gives Weber’s constant $\Delta I/I$ (Fig. 4). This is the efficient-coding account of Weber–Fechner [13], recovered here from the same surprisal currency: the currency that drives subjective time also predicts the classic intensity law, with no extra assumption beyond the prior.

Clinical signatures: a passage/estimation dissociation. Conditions that alter the predictive machinery should distort subjective time in specific, falsifiable directions. Driving the same online learner with three altered parameters—engagement ρ (the fraction of events registered), arousal (pacemaker rate), and precision—and reading out both the momentary passage (the surprisal rate) and an explicit attentional clock reproduces the directional signs reported across three clinical literatures (Fig. 5). Depression, modeled as reduced engagement, slows the felt *passage* of time while leaving explicit duration *estimation* intact—the dissociation found meta-analytically [14] ($g = 0.66$ slower passage; estimation tasks null). A single accumulated-surprisal clock cannot produce this dissociation, but the two readouts can: ρ gates the passage accumulator, not the explicit clock. Heightened arousal (mania) speeds the internal clock, predicting the shorter interval reproductions of manic relative to depressed patients [15]. Aberrant, unstable precision (aberrant salience) degrades timing *precision* far more than accuracy, matching the schizophrenia meta-analysis [16] (variability $g > 1$; accuracy effects small) under the aberrant-salience account of psychosis [17]. These are consistency results rather than new data, and the separate attentional clock is the standard scalar-timing account; the contribution is that one currency plus an attentional clock predicts the *pattern* of distortions—and, in depression, the dissociation—across conditions, falsifiable per condition by its sign.

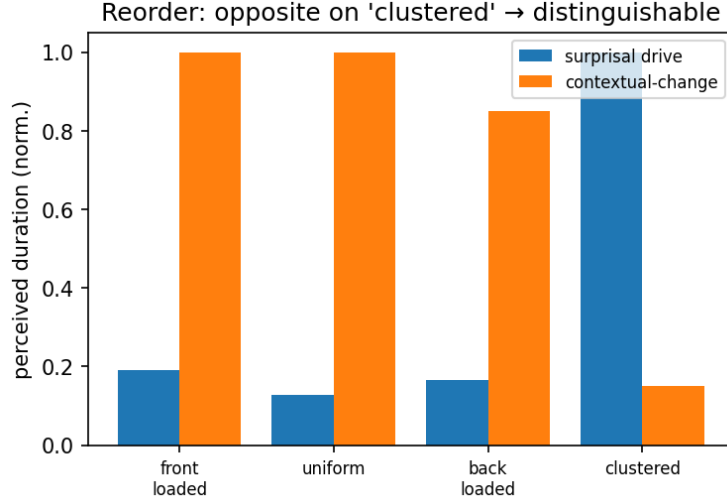


Figure 3: Reorder: the surprisal drive and a contextual-change account predict opposite signs on clustered arrangements, so the contrast both defines surprisal’s fingerprint and separates it from its nearest rival.

4 Embodied agents: an emergent convergence asymmetry

This section is a method as much as a result: *computational psychophysics of time*—synthetic subjects whose time-perception signatures emerge as byproducts of prediction, with no human-data bottleneck. The method has two design rules, both enforced here. The observer is *passive*: the agents optimize their own objective and never read the surprisal, so any time signature is a byproduct, not a built-in. And non-circularity is *enforced*: a probed and an unprobed agent sharing a seed follow identical trajectories, so attaching the readout cannot perturb what it measures. Within those rules the drive is fixed in advance and the subject is run, not fit.

The predictions above treat surprisal as a closed-form functional we write down. A stronger test is whether the same signatures *emerge* when the surprisal readout merely observes an agent built for a different purpose. We attach the identical online learner as a *passive* observer to autonomous programs in a deterministic 3D survival simulation: the agents optimize energy and survival via their own policies and never read the surprisal, so any time-perception signature is a byproduct, not a built-in. (A probed agent and an unprobed agent with the same seed follow identical trajectories, enforcing non-circularity.)

An identical event is more surprising later (back-heaviness). The central emergent result is one the closed-form reorder model of §3 cannot produce, because that model assigns each event a *fixed* surprisal independent of when it occurs. In a genuine learner the predictive variance shrinks as structure is absorbed— $pv_t = \sigma^2 + 1/\tau_t$ with $\tau_t = \tau_0 + t/\sigma^2$ —so the surprisal of a fixed deviation δ , $\frac{1}{2} \log(2\pi pv_t) + \frac{1}{2} \delta^2/pv_t$, *increases* with experience for any salient event ($\delta^2 > pv_t$) and saturates at $\frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2} \delta^2/\sigma^2$. The same identical event thus lands harder the later it occurs, against a more confident model (Fig. 6): the embodied agent reproduces this closed-form expectation, and the effect holds in 100% of conditions across seeds and a wide range of predictor priors. This refines the reorder prediction: the front-versus-back axis is not merely encoding-dependent noise but carries a systematic back-heavy component in any predictive learner, and the saturation is the same mechanism that produces the age ceiling (§3).

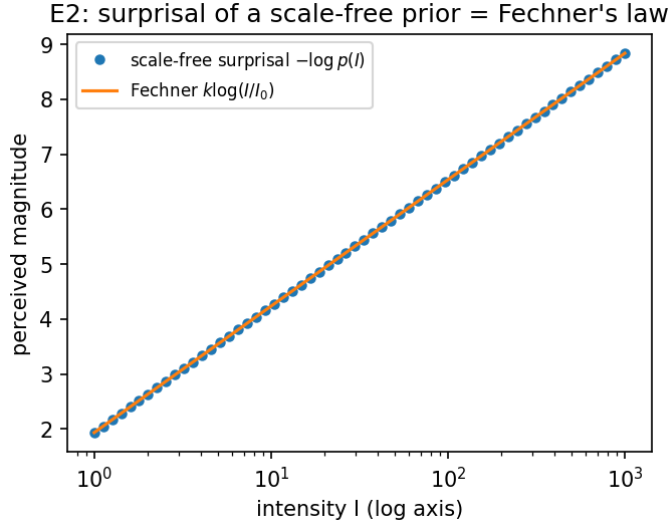


Figure 4: Scale-free surprisal equals Fechner’s log-magnitude; Weber’s law is its differential.

What survives embodiment, and what does not. Aggregating the same per-tick surprisal streams under the candidate readouts (Fig. 7), the clustered burst still reads long under the surprisal drive—consistent with the fingerprint of §3—and a leaky (recency-weighted) readout makes back-loaded arrangements longest. But the convergence asymmetry dominates all of them: back-loaded exceeds front-loaded across pure, leaky, and pacemaker readouts, so the abstract front/back *opposition* between recency and arousal accounts does not survive embodiment. We report two honest negatives rather than tuning them away. First, a forward (pacemaker) kernel does not flip the ordering to front-longest, because the surprisal substrate is itself back-heavy. Second, a crude embodied contextual-change readout did not reproduce clustered-*shortest* at any temporal resolution—the convergence asymmetry again dominated—so the clean clustered anti-correlation that motivates the decisive test (§6) remains an abstract-model prediction to be settled by the human experiment, not an embodied result. Relatedly, an environmental-volatility→felt-time dose-response is positive but *weak* on volatility measures computed independently of the surprisal stream (Spearman $\rho \approx 0.4$, seed-dependent; a naive axis derived from the surprisal stream itself inflates this to ≈ 0.8 and should not be trusted)—consistent with the engagement-not-capacity reading of the locality result below.

A content footprint in human data. Independently of the simulation, the surprisal account predicts that perceived duration is driven by stimulus *content*, not elapsed time alone. In the public prospective dataset of (author?) [7] (the model of (author?) [6]; 6,023 trials, 33 natural-scene videos), after regressing each estimate on the true duration, the *identity* of the video still explains residual variance: per-video bias spans -2.1 to $+1.8$ s (permutation $p = 0.001$), alongside the expected central-tendency bias (slope 0.74) and a counting-reduces-overestimation effect. Content biases felt duration beyond clock time, in humans. This is necessary but not sufficient—it is consistent with any content-driven account and re-confirms an established finding; the decisive, uniquely-surprisal test (whether our per-video surprisal predicts *which* videos are over-estimated) is blocked because the stimulus videos were never released.

These are synthetic subjects: the readout is the theory, so the contribution is a mechanistic existence proof and a sharp prediction—the closed-form, embodiment-robust back-heaviness—rather

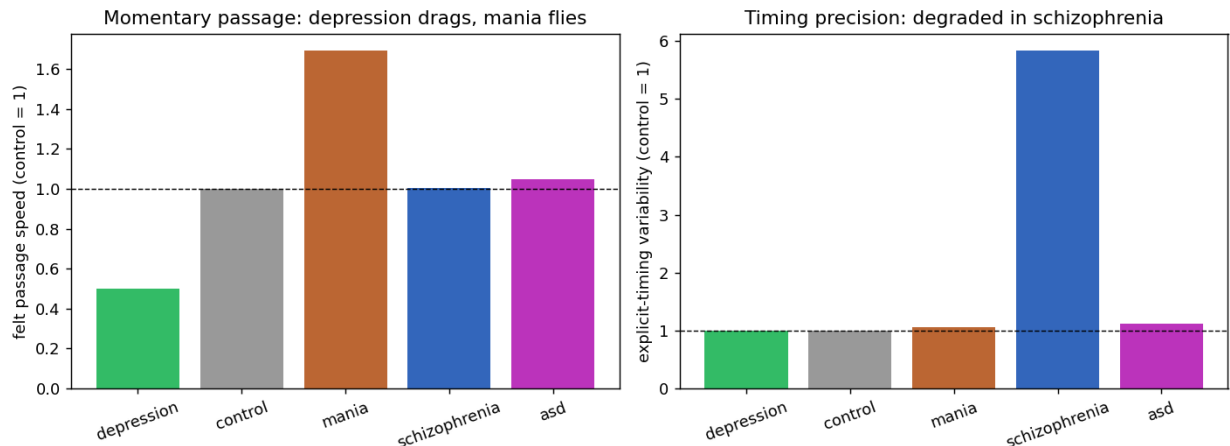


Figure 5: Predictive-processing clinical signatures (control = 1). Left: felt passage speed—reduced engagement (depression) drags, heightened arousal (mania) flies. Right: explicit-timing variability—degraded under aberrant precision (schizophrenia) while depression’s explicit timing is intact (the passage/estimation dissociation).

than human validation of the drive.

5 A first empirical test: locality slows felt time

The drive’s hard-to-observe quantity is the surprisal/novelty *rate*. We propose that the spatial extent a person inhabits—their *locality*—is an observable proxy: a smaller locality (confinement, routine) delivers less novelty per unit time. The momentary prediction is then that smaller locality *slows* felt time (monotony drags), the retrospective prediction that it *compresses* remembered duration (few memorable events).

We test the momentary arm in the open Blursday database [18], in which COVID-19 lockdowns acted as a natural manipulation of locality. Regressing the felt passage of time (a 0–100 visual-analog scale, 0 = very slow) on a self-reported subjective-confinement index, with age, session, and country covariates and country-clustered errors (Fig. 8), we find that greater confinement predicts a slower passage of time: $\beta = -1.11$ scale points per confinement-index unit ($p \approx 4 \times 10^{-12}$; $N = 2,376$ observations, 1,899 participants, 9 countries), about -17 points across the observed confinement range. The direction and magnitude match the prediction and the broader lockdown literature, in which boredom and inactivity—the locality→novelty pathway—mediate the effect.

Two sharper follow-ups, however, qualify this and partly contradict the *spatial* reading. (i) Replacing the subjective index with an *objective* transit-mobility index (Google) yields a null effect ($\beta = +0.016$, $p = 0.80$; null with or without session controls, and mobility is ~ 0.8 collinear with lockdown phase): the strong effect rides on *felt isolation*, not measured movement. (ii) The retrospective arm runs the wrong way: greater confinement *expands* remembered duration (log estimate/actual, $\beta = +0.004$, $p = 0.002$, $\sim +6\%$ across the range), the opposite of the predicted compression—though consistent with reports of time “stretching” during lockdown. So what survives is narrower than the spatial hypothesis: subjective confinement slows *momentary* felt time, robustly, but the objective-locality (state-counting) prediction of §2.4 is not supported and the retrospective prediction fails. In the $r = \rho \log W$ decomposition this points to the *engagement* term ρ (boredom/inactivity), not the capacity W , as the operative variable—a framework-consistent

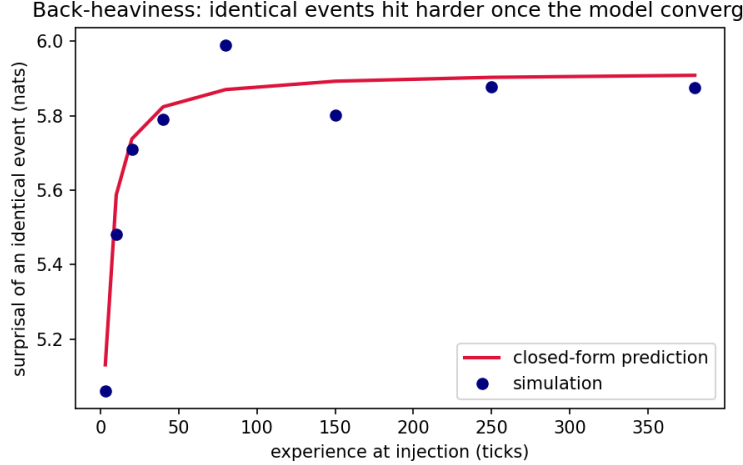


Figure 6: An identical event injected at increasing experience becomes more surprising and then plateaus; the embodied simulation (points) tracks the closed-form expectation (line). This convergence asymmetry is derivable in closed form and absent from a fixed-per-event accumulator.

reading, but one the spatial framing did not anticipate.

A regime the readout cannot reach. A companion simulation asks whether the model’s retrospective readout could even produce a confinement-driven distortion in this regime. It cannot: the confinement–distortion slope collapses from -0.73 under varied surprise to ≈ 0 once surprise is floored, and the interior distortion vanishes for a single undifferentiated interval, growing only once the interval carries several distinguishable segments. Lockdown confinement is plausibly both floored and undifferentiated, so the observed retrospective *expansion* is most consistent with an exogenous (emotional/anchoring) effect on which the surprisal readout is silent in that regime, rather than a refutation of the account. This does not settle the arm; it delimits, computationally, the conditions under which the account makes a retrospective prediction at all.

6 The decisive test (preregisterable)

The consistency checks above do not *adjudicate* surprisal against its nearest rival, because most accounts predict the same sign on most manipulations. One contrast does separate them. Under surprisal, a dense burst of events after a calm baseline is maximally informative and reads as the *longest* interval; under a contextual-change account, the same burst occupies the *fewest* distinct contexts and reads as the *shortest*. In simulation the two are essentially anti-correlated on this clustered contrast (normalized clustered-vs-dispersed score $+0.96$ surprisal vs -0.94 contextual-change; §3). The experiment is therefore a single, high-powered question.

Design. Within-subject; matched event sets (n events, fixed total, fixed interval endpoints) presented in four arrangements—front-loaded, uniform, back-loaded, and clustered (a dense mid-interval burst). DV: perceived interval duration (reproduction and a 2AFC replication). Stimulus information/change content is logged per trial (a model-based surprisal stream) as a covariate.

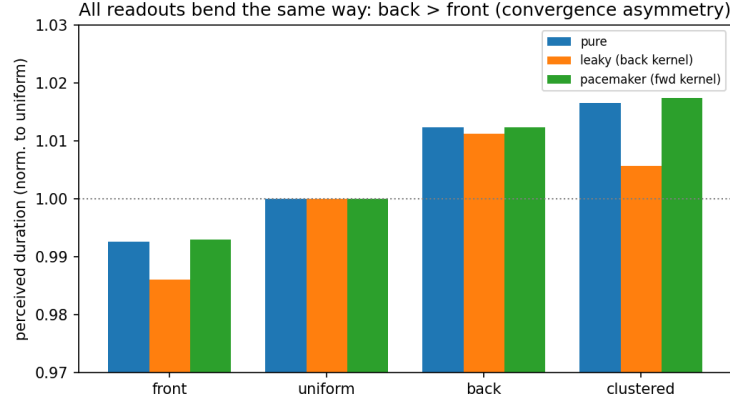


Figure 7: Perceived duration (normalized to uniform) under pure, leaky, and pacemaker readouts of the same embodied event streams. All show back > front—the convergence asymmetry—so the abstract front/back opposition does not survive embodiment; the clustered burst still reads long.

Preregistered predictions. (a) *Surprisal*: clustered > dispersed arrangements (clustered longest). (b) *Contextual-change*: clustered < dispersed (clustered shortest). (c) *Leaky-accumulator / pacemaker*: ordered front/back effects with clustered intermediate. The clustered-vs-dispersed *sign* is the decisive quantity; the front/back axis is not (it is encoding-dependent for surprisal, §3).

Decision rule & falsifier.

Clustered rated *longest*, effect scaling with the burst’s logged surprisal $\Rightarrow$ **surprisal**.
Clustered rated *shortest* $\Rightarrow$ **contextual-change**, and surprisal is *falsified*.
A null clustered contrast *undercuts both*.

Power for a medium within-subject effect needs only tens of participants; the design, the surprisal model, and exclusions are fixed in advance. The contrast is established here *in simulation* (+0.96 surprisal vs -0.94 contextual-change, §3); as §4 records, the clean anti-correlation remains an abstract-model prediction to be settled by this behavioural test—which we offer as the field’s decisive falsifier, not one we run. This is the experiment the paper is built to provoke: a model-agnostic contrast that returns a verdict, not a consistency check.

7 Discussion

A single currency—surprisal—supplies the path-distinguishing drive the formal note requires, and from it a connected set of predictions follows: a finite age ceiling (supported by lifespan data to 94), a clustered-event order effect that also separates surprisal from contextual-change, Weber–Fechner scaling, and an isolation effect on momentary felt time that we confirm directly. The program is falsifiable at each node, and it has already taken damage: locality tested by *objective* mobility is null, and the retrospective prediction is wrong-signed (§5), so the spatial state-counting reading is not supported—what holds is the narrower, affect-laden claim that felt isolation slows momentary time, plausibly via engagement rather than spatial capacity. The framework nonetheless survives where the popular alternatives (proportional time, coordinate-rate drives) fail, derives an age ceiling the lifespan data bear out, and makes a battery of still-open falsifiable predictions—chief among

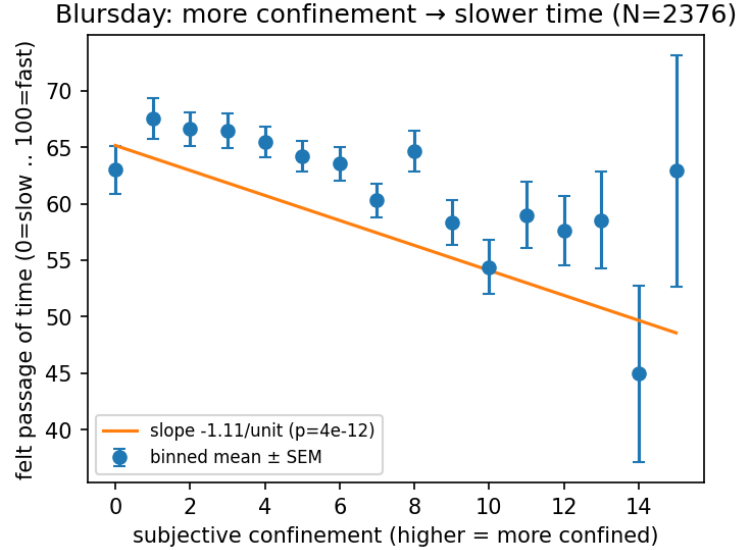


Figure 8: Blursday ($N = 2,376$): greater subjective confinement predicts a slower felt passage of time (binned means $\pm$ SEM; cluster-robust slope).

them the decisive clustered-burst contrast of §6 (the 85+ age tail and a within-person spatial-diversity test that separates capacity from engagement round out the set). We take this as warrant for surprisal as a candidate organizing principle—not a settled one—and as a sharpened target for the experiments that would decide it.

Stated plainly, the companion note’s exclusion of a faster clock leaves one picture standing: you cannot feel time itself, only what filled it. Perceived duration is the deposit experience lays down—a child’s year, dense with firsts, is a fat deposit and feels enormous; an adult’s year of reruns is a thin one, and “where did the year go?” The corollary is structural, not motivational: you cannot slow the clock, only fatten the deposit, so novelty density—not rate—is the only lever the theory leaves on felt time.

8 Methods

All predictions are implemented as additive, tested simulation modules (surprisal drive, reorder, age, identifiability, memory-kernel, Weber–Fechner) with a model-selection solver (variable-projection fits, AIC/BIC, bootstrap confidence). The age data are figure-digitized from (author?) [11] (shape, not calibrated ratio). The Blursday analysis uses the openly licensed data [18]; confinement indices derive from a UCLA-loneliness subset [19] and the analysis is a linear model with country-clustered standard errors. Code and figure generators regenerate every in-repo result; the embodied-agent simulation (§4) runs in a separate companion codebase (see Data and code availability).

Data and code availability

All simulation and analysis code and figure generators for the abstract-model and empirical results are openly available in the project repository. The embodied-agent results (Fig. 6, Fig. 7) were produced by a separate 3D-simulation companion codebase in which the surprisal readout is a passive observer; that codebase is available from the authors on request. The Blursday database

is openly available (CC BY) at <https://osf.io/359qm/>. Run-artifact provenance follows the companion note’s reference-manifest scheme.

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